

ASP Problem Set 1

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September 22, 2026

1. For each element of $C = AB$,

$$C_{ij} = \sum_{k=1}^N A_{ik} B_{kj}.$$

Computing one element requires N multiplications and $N - 1$ additions, so the total number of operations for one element is $N + (N - 1) = 2N - 1$, which is $O(N)$.

Since there are N^2 elements in C , the total number of operations is $(2N - 1)N^2 = 2N^3 - N^2$. Therefore, the computational complexity is

$$O(N^3).$$

2. (a) For the leave-one-out estimator,

$$T_{(-i)} = \frac{1}{n-1} \sum_{j \neq i} X_j = \frac{1}{n-1} \left(\sum_{j=1}^n X_j - X_i \right) = \frac{1}{n-1} (n\bar{X} - X_i).$$

- (b) The average of the leave-one-out estimators is

$$\bar{T}_{(-)} = \frac{1}{n} \sum_{i=1}^n T_{(-i)} = \frac{1}{n} \sum_{i=1}^n \frac{1}{n-1} (n\bar{X} - X_i) = \frac{1}{n-1} \left(n\bar{X} - \frac{1}{n} \sum_{i=1}^n X_i \right) = \frac{1}{n-1} (n\bar{X} - \bar{X}) = \bar{X}.$$

- (c) The leave-one-out estimate of the bias is

$$\widehat{\text{Bias}}_{\text{loo}} = (n-1) (\bar{T}_{(-)} - T_n) = (n-1) (\bar{X} - \bar{X}) = 0.$$

- (d) The bias-corrected estimator is

$$T_{\text{loo}} = T_n - \widehat{\text{Bias}}_{\text{loo}} = T_n.$$

For each i , X_i is random and independent of the other variables. Therefore, the leave-one-out sample has the same expectation as the original sample, so

$$E(T_{(-i)}) = E(T_n) = E(\bar{X}).$$

3. (a) We know that $X_1, \dots, X_n \sim \text{Unif}[0, 1]$, so

$$F(x) = \begin{cases} 0, & x < 0, \\ x, & 0 \leq x \leq 1, \\ 1, & x > 1. \end{cases}$$

Since $T_n = \min\{X_1, \dots, X_n\}$ and $X_1, \dots, X_n$ are i.i.d.,

$$\Pr(nT_n > x) = \Pr\left(T_n > \frac{x}{n}\right) = \Pr\left(X_1 > \frac{x}{n}, \dots, X_n > \frac{x}{n}\right) = \prod_{i=1}^n \Pr\left(X_i > \frac{x}{n}\right) = \left(1 - \frac{x}{n}\right)^n.$$

Since

$$\lim_{n \rightarrow \infty} \left(1 - \frac{x}{n}\right)^n = e^{-x},$$

the CDF of nT_n satisfies $G(x) = 1 - \Pr(nT_n > x) \rightarrow 1 - e^{-x}$ as $n \rightarrow \infty$. Therefore,

$$nT_n \xrightarrow{d} \text{Exp}(1).$$

(b) For each i , $\Pr(X_i^* \neq T_n) = 1 - \frac{1}{n}$. Therefore,

$$\Pr(X_1^* \neq T_n, \dots, X_n^* \neq T_n) = \left(1 - \frac{1}{n}\right)^n.$$

Since

$$\lim_{n \rightarrow \infty} \left(1 - \frac{1}{n}\right)^n = e^{-1},$$

The probability that the bootstrap sample does not contain T_n converges to e^{-1} as $n \rightarrow \infty$.

(c) From (b), we can get

$$\lim_{n \rightarrow \infty} \Pr(T_n^* = T_n) = 1 - e^{-1}.$$

This implies that T_n^* does not converge to T_n as $n \rightarrow \infty$ with probability greater than 0. Therefore, the nonparametric bootstrap is not valid in this setting.

4. (a) Let $Z_i = 1$ denote tea first in cup i , and let $Z_i = 0$ denote milk first in cup i . Let $Y_i(z)$ denote Muriel's classification under assignment z .

The sharp null hypothesis is

$$H_0 : Y_i(1) = Y_i(0) \quad \text{for every } i = 1, \dots, 8.$$

(b) We know that there are four tea-first cups and four milk-first cups among eight cups, so the number of possible assignments is

$$\binom{8}{4} = \frac{8!}{4!4!} = 70.$$

(c) Let k denote the number of correctly classified tea-first cups, where $k \in \{0, 1, 2, 3, 4\}$. By symmetry, k milk-first cups are also classified correctly. Therefore, the total number of correctly classified cups is $S = 2k$.

For each k , the number of corresponding assignments is $\binom{4}{k}^2$.

k	S	Number of assignments
0	0	$\binom{4}{0}^2 = 1$
1	2	$\binom{4}{1}^2 = 16$
2	4	$\binom{4}{2}^2 = 36$
3	6	$\binom{4}{3}^2 = 16$
4	8	$\binom{4}{4}^2 = 1$

(d) The exact one-sided permutation p -value is

$$p = \Pr(S \geq S_{\text{obs}} \mid H_0) = \Pr(S = 8 \mid H_0) = \frac{1}{70} \approx 0.0143.$$

(e) Since $p = 0.0143 < 0.05$, the probability that Muriel classified all eight cups correctly under randomized assignment, if she could not distinguish between the cups, is only approximately 0.0143. Therefore, I would reject the null hypothesis at the 5% significance level.

5. (e) From (d), we know that $p\text{-value} = 0.00031 < 0.05$, so we reject the null hypothesis at the 5% significance level. In this context, the observed fact that this participant successfully classified 67 cups if she could not distinguish between cups is less than 5%, which implies that she is likely to be able to distinguish cups.